

# AK High-Dimensional Projection Structural Theory

## Appendix TB-v19

### Toy Bridge Theorem Pack

Exact Tower Artifacts, Finite-Length Stabilization,  
Terminal-Generic Defect Calculus, and Metric-Safe Audit

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Evidence status: Mixed proved / conditional / operational / non-claim  
Conformance: v19 mathematical conformance; CR-r3 promotion pending  
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**Normative boundary.** This appendix proves bridge statements only for deliberately small toy source categories and explicitly declared Core-readable realizations. It does not prove any arithmetic, PDE, geometric, categorical-equivalence, or cryptographic theorem. It consumes the v19.0.0 Main, Foundation A–G, CR-v19.0.0-r2 governance, and CM-v19.0.0-r2 micro-theorem pack. Formal promotion of the local TB UUIDs into CR-v19.0.0-r3 is a separate immutable governance revision. Artifact-closure patch p1 changes only the complete human-readable register display, portable font policy, and build/package identities; it does not change any theorem statement, hypothesis, UUID, or verdict.

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# 1 Scope, Dependency Baseline, and Theorem Boundary

This appendix replaces the legacy v18.3 Candidate TB document. It is not a string-level migration. Its mathematical architecture is rebuilt around the repaired v19 comparison calculus, exact tower artifacts, terminal-generic Type IV semantics, typed failure routes, and separation of metric and diagnostic quantities.

**Principle 1.1** (Toy bridge sovereignty). *A toy bridge theorem may transport a proved source-side toy condition to a Core-visible conclusion only after the source category, realization route, monitored scope, comparison artifact, and required evidence have been declared. The conclusion is limited to the toy source semantics actually proved.*

|                          |                                          |
|--------------------------|------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-001 |
| <b>Role / status</b>     | operational_policy / operational         |
| <b>Source bindings</b>   | CR-G-001, CR-G-002, CR-G-004, CR-G-007   |
| <b>Comparison policy</b> | not_applicable; runtime modes: none      |
| <b>Failure routes</b>    | FR-MISSING, FR-T5, FR-NONVERDICT         |

**Principle 1.2** (No external-domain upgrade). *No theorem in Appendix TB-v19 proves BSD, ABC, RH, an Iwasawa main conjecture, Navier–Stokes regularity, a geometric equivalence, mirror symmetry, a Fukaya-categorical equivalence, or a cryptographic security theorem. A toy theorem can be consumed by a later Bridge Program, but it cannot substitute for a problem-specific Interface or Bridge Theorem.*

**Definition 1.3** (Immutable dependency baseline). *The normative dependency order for this appendix is:*

1. AK-HPDST v19.0.0 Main Chapters 1–8;
2. Foundation Appendices A–G;
3. CR-v19.0.0-r2 as the currently available claim-governance snapshot;
4. CM-v19.0.0-r2 as the direct theorem source for reusable micro-theorems;
5. this Appendix TB-v19.0.0-r1 and its local machine-readable register.

Where CR-r2 still labels CM as pending, this appendix binds directly to the immutable CM-r2 artifact and records CR-r3 promotion as a governance dependency rather than fabricating a completed promotion.

| Artifact ID        | Artifact / availability                        | SHA-256 digest     |
|--------------------|------------------------------------------------|--------------------|
| SRC-MAIN-PDF       | AK-HPDST v19.0.0 Main; verified dependency     | 7d926a21...ce5e750 |
| SRC-FOUNDATION-PDF | Foundation Appendices A–G; verified dependency | 12489aaf...6c982   |
| SRC-CR-R2-PDF      | CR-v19.0.0-r2; verified dependency             | bf9895a7...800e5   |
| SRC-CM-R2-PDF      | CM-v19.0.0-r2; verified dependency             | a6c33c8e...5b65f   |

| Artifact ID           | Artifact / availability                                                                                | SHA-256 digest     |
|-----------------------|--------------------------------------------------------------------------------------------------------|--------------------|
| SRC-TB-<br>LEGACY-PDF | Historical v18.3 PDF identity;<br>digest-only repair ancestry, not bound to<br>the current TB basename | da47f9db...6b3a86f |
| SRC-TB-<br>LEGACY-TEX | Historical v18.3 source text; bundled<br>under an immutable legacy filename                            | 9f0bc5bb...0e86db3 |

**Remark 1.4** (Digest display and immutable identity). *The shortened digests above are navigational display strings only. The complete 64-hex SHA-256 values, availability flags, and immutable basenames are normative in the machine-readable register and package manifest. The current canonical TB PDF is never identified by the historical legacy basename.*

**Guard-rail.** Registration is not proof. A complete manifest is not proof. A successful XeLaTeX build is not proof. Each theorem below is valid only at the strength of its displayed hypotheses and source bindings.

## 2 Standing Repaired Core Regime

Fix a field  $k$ . Work in the constructible one-parameter persistence regime

$$\text{Pers}_k^{\text{cons}} \subset [(\mathbb{R}, \leq), \text{Vect}_k]$$

and with finite-type filtered objects in  $\text{FiltCh}(k)$  whenever a filtered realization is used.

For a monitored degree  $i$ , right-open window  $W$ , and threshold  $\tau > 0$ , set

$$B_{i,W}(F) := W_W P_i(F), \quad \text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(B_{i,W}(F)).$$

Windowing precedes hard deletion. Threshold clearance is measured on  $B_{i,W}(F)$  before deletion.

**Definition 2.1** (Canonical comparison modes). *Every comparison consumed by a TB theorem resolves to exactly one runtime mode:*

*exact,      metric\_gap\_certified,      not\_compared.*

*The mode **not\_compared** is permitted only when the theorem consumes no comparison inference. It cannot discharge an invoked tower, representative, overlap, or transport obligation.*

|                          |                                                             |
|--------------------------|-------------------------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-002                    |
| <b>Role / status</b>     | operational_policy / operational                            |
| <b>Source bindings</b>   | CR-CH1-004, CM-v19-015                                      |
| <b>Comparison policy</b> | runtime_resolved; exact, metric_gap_certified, not_compared |
| <b>Failure routes</b>    | FR-MISSING, FR-T3, FR-T5                                    |

**Definition 2.2** (Typed Type IV scope). *For each invoked monitored triple  $(i, W, \tau)$ , an actual persistence morphism*

$$\phi_{i,W,\tau} : M_{i,W,\tau} \longrightarrow N_{i,W,\tau}$$

*is supplied. When source, target, kernel, and cokernel are terminal-tame on the declared scope, define*

$$\mu_{i,W,\tau} := \text{tgd}_W(\ker \phi_{i,W,\tau}), \quad \nu_{i,W,\tau} := \text{tgd}_W(\text{coker } \phi_{i,W,\tau}).$$

For a declared finite monitored set  $I_{\text{mon}}$ ,

$$\mu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \mu_{i,W,\tau}, \quad \nu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \nu_{i,W,\tau}.$$

The corresponding specialized status is exactly one of

`not_invoked`, `undefined`, `certified_zero`, `obstructed`.

No numerical pair is imputed in the first two cases.

### 3 Toy Source Categories and Bridge Data

**Definition 3.1** (Toy source category). *A toy source category is a declared small mathematical category or indexed diagram whose objects, morphisms, exact sequences, and colimits used by the bridge can be checked directly. The principal examples are:*

1.  $\text{ToyVect}_k$ : finite-dimensional  $k$ -vector spaces;
2.  $\text{ToyMod}_R$ : finite-length left modules over an Artinian  $k$ -algebra  $R$ ;
3. finite barcode-profile diagrams realized as actual morphisms in  $\text{Pers}_k^{\text{cons}}$ ;
4. finite direct sums, composable chains, and short exact comparison diagrams built from the preceding objects.

**Definition 3.2** (Toy bridge datum). *A toy bridge datum is a tuple*

$$\mathfrak{T} = (\mathcal{T}, X_{\bullet}, X_{\infty}, \mathcal{R}_{\text{Toy}}, I_{\text{mon}}, W, \tau, Q_{\bullet}, Q_{\infty}, \phi, \mathcal{E}, \mathcal{A}),$$

where:

1.  $\mathcal{T}$  is the toy source category;
2.  $X_{\bullet}$  is a directed toy tower and  $X_{\infty}$  is its declared apex;
3.  $\mathcal{R}_{\text{Toy}}$  is the declared realization route;
4.  $(I_{\text{mon}}, W, \tau)$  is the immutable monitored scope;
5.  $Q_n^{i,W,\tau}$  and  $Q_{\infty}^{i,W,\tau}$  are repaired Core-readable profiles;
6.  $\phi$  is the actual tower comparison artifact when Type IV is invoked;
7.  $\mathcal{E}$  is the evidence bundle;
8.  $\mathcal{A}$  is the immutable artifact and provenance bundle.

**Definition 3.3** (Core-readable toy realization). *A realization route  $\mathcal{R}_{\text{Toy}}$  is Core-readable on the declared scope when it supplies actual objects and morphisms in  $\text{FiltCh}(k)$  or  $\text{Pers}_k^{\text{cons}}$ , repaired evaluations exist, constructibility is verified, and every invoked representative, eligibility, tower, or metric obligation is supplied as a separately typed record.*

**Proposition 3.4** (Finite barcode readability). *Let  $Q$  be a finite barcode profile with declared endpoint convention, right-open window  $W$ , and threshold  $\tau > 0$ . Then  $T_{\tau}^{\text{bar}}(W_W Q)$  is again a finite barcode profile. A tower of such profiles is Core-readable only after its transition data are*

realized as actual persistence morphisms or are explicitly restricted to object-level, non-Type-IV use.

*Proof.* Window restriction clips finitely many intervals and discards empty intersections. Hard deletion removes a submultiset of the resulting finite multiset. Thus the output is finite. The final qualification is necessary because object-level barcodes alone do not manufacture the actual morphism required by Type IV diagnostics.  $\square$

|                          |                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-003                                                   |
| <b>Role / status</b>     | toy_bridge_theorem / proved                                                                |
| <b>Source bindings</b>   | CM-v19-008, CM-v19-009, CM-v19-022                                                         |
| <b>Comparison policy</b> | not_applicable at object level; exact or metric mode required when comparisons are invoked |
| <b>Failure routes</b>    | FR-MISSING, FR-T3, FR-T5                                                                   |

## 4 Exact Tower Artifacts and Eventual Constancy

**Definition 4.1** (Exact toy tower artifact). *For a repaired profile tower*

$$Q_0 \longrightarrow Q_1 \longrightarrow Q_2 \longrightarrow \dots$$

*with declared apex  $Q_\infty$ , an exact toy tower artifact is an actual morphism*

$$\phi^{\text{stab}} : \operatorname{colim}_n Q_n \longrightarrow Q_\infty$$

*constructed from a declared colimit and an artifact-backed apex agreement. Its runtime comparison mode is **exact**.*

**Theorem 4.2** (Eventual-constancy toy bridge). *Let  $Q_0 \rightarrow Q_1 \rightarrow \dots$  be a repaired profile tower in a category admitting the declared colimit. Suppose:*

1. *there exists  $N$  such that every transition  $Q_n \rightarrow Q_{n+1}$  is an isomorphism for  $n \geq N$ ;*
2. *an isomorphism  $a : Q_N \xrightarrow{\sim} Q_\infty$  is declared;*
3.  *$\phi^{\text{stab}}$  is the colimit-to-apex morphism induced by the stabilized diagram and  $a$ ;*
4. *when Type IV is invoked, the source, target, and defect profiles are terminal-tame on the same monitored scope.*

*Then  $\phi^{\text{stab}}$  is an isomorphism. The Type IV status is **certified\_zero** and*

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0).$$

*Proof.* The colimit of an eventually constant diagram is canonically isomorphic to the stabilized value  $Q_N$ . Composition with  $a$  gives an isomorphism to  $Q_\infty$ . Kernel and cokernel vanish. Terminal-generic dimensions therefore vanish in every monitored degree.  $\square$

|                          |                                              |
|--------------------------|----------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-004     |
| <b>Role / status</b>     | conditional_toy_bridge_theorem / conditional |
| <b>Source bindings</b>   | CM-v19-031                                   |
| <b>Comparison policy</b> | fixed_exact; runtime mode: exact             |
| <b>Failure routes</b>    | FR-MISSING, FR-T3, FR-T4, FR-T5              |

**Corollary 4.3** (Finite barcode-profile stabilization). *Suppose a tower of finite barcode profiles is realized as an actual tower in  $\text{Pers}_k^{\text{cons}}$ , becomes isomorphic after a finite stage, admits the declared colimit, and satisfies apex agreement. Then the induced exact tower artifact is an isomorphism and its valid terminal-generic Type IV diagnostics vanish.*

*Proof.* Apply the eventual-constancy toy bridge to the realized persistence tower. The realization and actual-morphism hypotheses are essential; a sequence of multisets without morphisms is insufficient.  $\square$

|                          |                                              |
|--------------------------|----------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-005     |
| <b>Role / status</b>     | conditional_toy_bridge_theorem / conditional |
| <b>Source bindings</b>   | TB-v19-003, TB-v19-004                       |
| <b>Comparison policy</b> | fixed_exact; runtime mode: exact             |
| <b>Failure routes</b>    | FR-MISSING, FR-T3, FR-T4, FR-T5              |

**Theorem 4.4** (Exact-conjugacy invariance of toy diagnostics). *Let  $\phi : M \rightarrow N$  and  $\phi' : M' \rightarrow N'$  be actual toy comparison artifacts. Suppose there are isomorphisms  $\alpha : M \xrightarrow{\sim} M'$  and  $\beta : N \xrightarrow{\sim} N'$  with*

$$\beta \circ \phi = \phi' \circ \alpha.$$

*Then  $\ker \phi \cong \ker \phi'$  and  $\text{coker } \phi \cong \text{coker } \phi'$ . On a common valid terminal-tame scope, all degreewise and aggregated  $\mu/\nu$  diagnostics agree.*

*Proof.* The source isomorphism restricts to kernels and the target isomorphism descends to cokernels. Terminal-generic dimension is invariant under persistence-module isomorphism.  $\square$

|                          |                                          |
|--------------------------|------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-006 |
| <b>Role / status</b>     | toy_bridge_theorem / proved              |
| <b>Source bindings</b>   | CM-v19-033                               |
| <b>Comparison policy</b> | fixed_exact; runtime mode: exact         |
| <b>Failure routes</b>    | FR-MISSING, FR-T3, FR-T5                 |

## 5 Finite-Dimensional Vector-Space Toy Towers

**Theorem 5.1** (Monomorphism stabilization in finite dimension). *Let*

$$V_0 \xrightarrow{f_0} V_1 \xrightarrow{f_1} V_2 \rightarrow \dots$$

*be a tower in  $\text{ToyVect}_k$ . If every  $f_n$  is injective and there is a finite bound  $D$  with  $\dim_k V_n \leq D$  for all  $n$ , then the tower is eventually a tower of isomorphisms.*

*Proof.* The integer sequence  $\dim_k V_n$  is nondecreasing and bounded by  $D$ , hence eventually constant. An injective linear map between finite-dimensional vector spaces of equal dimension is an isomorphism.  $\square$

**Theorem 5.2** (Epimorphism stabilization in finite dimension). *Let  $V_0 \rightarrow V_1 \rightarrow \dots$  be a tower in  $\text{ToyVect}_k$  whose transition maps are all surjective. Then it is eventually a tower of isomorphisms.*

*Proof.* The nonnegative integer sequence  $\dim_k V_n$  is nonincreasing, hence eventually constant. A surjective map between finite-dimensional spaces of equal dimension is an isomorphism.  $\square$

|                          |                                          |
|--------------------------|------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-007 |
| <b>Role / status</b>     | toy_source_theorem / proved              |
| <b>Source bindings</b>   | finite-dimensional rank-nullity          |
| <b>Comparison policy</b> | not_applicable                           |
| <b>Failure routes</b>    | FR-MISSING, FR-T5                        |

**Corollary 5.3** (Vector-space stabilization bridge). *Let a vector-space toy tower satisfy either preceding stabilization theorem. Suppose  $\mathcal{R}_{\text{Toy}}$  sends the eventual source isomorphisms to repaired profile isomorphisms, the repaired tower admits the declared colimit, and apex agreement and terminal-tameness records are supplied. Then its exact tower artifact has status **certified\_zero**.*

*Proof.* The source theorem gives eventual isomorphism. The declared realization transfers this to eventual constancy of repaired profiles. Apply the eventual-constancy toy bridge.  $\square$

|                          |                                              |
|--------------------------|----------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-008     |
| <b>Role / status</b>     | conditional_toy_bridge_theorem / conditional |
| <b>Source bindings</b>   | TB-v19-004, TB-v19-007                       |
| <b>Comparison policy</b> | fixed_exact; runtime mode: exact             |
| <b>Failure routes</b>    | FR-MISSING, FR-T3, FR-T4, FR-T5              |

## 6 Finite-Length Module Toy Towers

Let  $R$  be an Artinian  $k$ -algebra and  $\text{len}_R(-)$  the composition length.

**Theorem 6.1** (Monomorphism stabilization in finite length). *Let*

$$M_0 \hookrightarrow M_1 \hookrightarrow M_2 \hookrightarrow \dots$$

*be a tower of finite-length left  $R$ -modules. If  $\text{len}_R(M_n) \leq L$  for a fixed finite  $L$ , then every transition is an isomorphism after some finite stage.*

*Proof.* The length sequence is nondecreasing and bounded in  $\mathbb{N}$ , hence eventually constant. In a short exact sequence

$$0 \rightarrow M_n \rightarrow M_{n+1} \rightarrow C_n \rightarrow 0,$$

length additivity gives  $\text{len}_R(C_n) = \text{len}_R(M_{n+1}) - \text{len}_R(M_n)$ . At a constant stage this is zero, so  $C_n = 0$  and the transition is an isomorphism.  $\square$

**Theorem 6.2** (Epimorphism stabilization in finite length). *Let  $M_0 \twoheadrightarrow M_1 \twoheadrightarrow \dots$  be a tower of finite-length left  $R$ -modules. Then it is eventually a tower of isomorphisms.*



*Proof.* The nonnegative integer sequence  $\text{len}_R(M_n)$  is nonincreasing and eventually constant. Length additivity for the kernel of a surjection then forces the kernel to vanish.  $\square$

|                        |                                          |
|------------------------|------------------------------------------|
| <b>TB UID</b>          | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-009 |
| <b>Role / status</b>   | toy_source_theorem / proved              |
| <b>Source bindings</b> | Jordan–Hölder length additivity          |
| <b>Comparison</b>      | not_applicable                           |
| <b>policy</b>          |                                          |
| <b>Failure routes</b>  | FR-MISSING, FR-T5                        |

**Corollary 6.3** (Finite-length module stabilization bridge). *Suppose a finite-length module tower satisfies either stabilization theorem,  $\mathcal{R}_{\text{Toy}}$  sends eventual module isomorphisms to repaired profile isomorphisms, and the colimit, apex agreement, exact artifact, and terminal-tameness records exist. Then the monitored Type IV status is **certified\_zero**.*

*Proof.* Apply the source stabilization theorem and then the eventual-constancy toy bridge.  $\square$

|                        |                                              |
|------------------------|----------------------------------------------|
| <b>TB UID</b>          | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-010     |
| <b>Role / status</b>   | conditional_toy_bridge_theorem / conditional |
| <b>Source bindings</b> | TB-v19-004, TB-v19-009                       |
| <b>Comparison</b>      | fixed_exact; runtime mode: exact             |
| <b>policy</b>          |                                              |
| <b>Failure routes</b>  | FR-MISSING, FR-T3, FR-T4, FR-T5              |

**Remark 6.4** (Iwasawa boundary). *A finite-length module toy tower is not an Iwasawa tower. No passage from finite-length stabilization to an Iwasawa control theorem, information-preserving inverse/direct limit, characteristic ideal, or  $p$ -adic  $L$ -function statement is claimed.*

## 7 Eligible Degree-One Toy Edge

**Theorem 7.1** (One-way eligible toy  $\text{PH}_1$ -to- $\text{Ext}^1$  bridge). *Let a toy source object  $X$  admit a Core-readable filtered realization  $F = \mathcal{R}_{\text{Toy}}(X)$ . Suppose the complete eligible regime is verified: an admissible representative  $C_{\tau, W}F$  exists, its monitored profiles agree with the repaired evaluations, its realization*

$$A := \mathcal{R}_{\text{Core}}(C_{\tau, W}F) \in D^{[-1, 0]}(k\text{-mod})$$

*is defined, and the degree-one edge-realization record identifies*

$$H^{-1}(A) \cong H_1^{\tau, W}(F).$$

*If*

$$\text{Eval}_{1, W, \tau}(F) = 0,$$

*then*

$$\text{Ext}^1(A, k) = 0.$$

*Proof.* This is the v19 conditional one-way repaired  $\text{PH}_1$ -to- $\text{Ext}^1$  bridge applied to the declared toy realization. The toy origin does not weaken any eligibility hypothesis.  $\square$

|                          |                                                                     |
|--------------------------|---------------------------------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-011                            |
| <b>Role / status</b>     | conditional_toy_bridge_theorem / conditional                        |
| <b>Source bindings</b>   | CM-v19-019, CM-v19-020                                              |
| <b>Comparison policy</b> | not_compared unless a separate representative comparison is invoked |
| <b>Failure routes</b>    | FR-MISSING, FR-T2, FR-T3, FR-T5                                     |

**Theorem 7.2** (Representative invariance of the eligible toy discharge). *Under the hypotheses of the preceding theorem, suppose  $A$  and  $A'$  are two eligible realized representatives connected by an artifact-backed isomorphism compatible with the degree-one edge records. Then*

$$\text{Ext}^1(A, k) = 0 \iff \text{Ext}^1(A', k) = 0.$$

*Proof.* Precomposition by the isomorphism induces an isomorphism of the corresponding  $\text{Ext}^1$  groups. Compatibility of edge records ensures that both groups discharge the same repaired degree-one clause.  $\square$

|                          |                                              |
|--------------------------|----------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-012     |
| <b>Role / status</b>     | conditional_toy_bridge_theorem / conditional |
| <b>Source bindings</b>   | CM-v19-032                                   |
| <b>Comparison policy</b> | fixed_exact; runtime mode: exact             |
| <b>Failure routes</b>    | FR-MISSING, FR-T3, FR-T5                     |

**Guard-rail.** No reverse implication and no global equivalence  $\text{PH}_1 \cong \text{Ext}^1$  is proved here. The theorem is one-way and eligibility-dependent.

## 8 Terminal-Generic Defect Calculus

**Theorem 8.1** (Terminal-generic index identity). *Let  $\phi : M \rightarrow N$  be an actual morphism between terminal-tame persistence modules with finite-dimensional stabilized tails. Then*

$$\mu(\phi) - \nu(\phi) = \text{tgdim}(M) - \text{tgdim}(N).$$

*If the tail dimensions agree, then  $\mu(\phi) = \nu(\phi)$ .*

*Proof.* On a common stabilized tail,  $\phi$  is a finite-dimensional linear map. Rank-nullity gives the displayed identity. The stabilized dimensions are the terminal-generic dimensions.  $\square$

|                          |                                          |
|--------------------------|------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-013 |
| <b>Role / status</b>     | toy_bridge_theorem / proved              |
| <b>Source bindings</b>   | CM-v19-034                               |
| <b>Comparison policy</b> | not_applicable                           |
| <b>Failure routes</b>    | FR-MISSING, FR-T4, FR-T5                 |

**Theorem 8.2** (Finite direct-sum additivity). *For terminal-tame actual morphisms  $\phi_j : M_j \rightarrow N_j$ ,  $1 \leq j \leq m$ ,*

$$\mu\left(\bigoplus_{j=1}^m \phi_j\right) = \sum_{j=1}^m \mu(\phi_j), \quad \nu\left(\bigoplus_{j=1}^m \phi_j\right) = \sum_{j=1}^m \nu(\phi_j).$$

Consequently, a finite direct sum is terminal-generically clean exactly when every summand is clean.

*Proof.* Kernels and cokernels commute with finite biproducts, and terminal-generic dimension is additive on finite direct sums. Nonnegativity gives the “exactly when” statement.  $\square$

|                          |                                          |
|--------------------------|------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-014 |
| <b>Role / status</b>     | toy_bridge_theorem / proved              |
| <b>Source bindings</b>   | CM-v19-035                               |
| <b>Comparison policy</b> | not_applicable                           |
| <b>Failure routes</b>    | FR-MISSING, FR-T4, FR-T5                 |

**Theorem 8.3** (Composition-controlled toy bridge). *Let*

$$M \xrightarrow{f} N \xrightarrow{g} P$$

*be terminal-tame actual morphisms on a common terminal scope. Then*

$$\mu(gf) \leq \mu(f) + \mu(g), \quad \nu(gf) \leq \nu(f) + \nu(g).$$

*In particular, two clean comparisons have a clean composite.*

*Proof.* On the stabilized tail, the kernel inequality follows from the map  $\ker(gf) \rightarrow \ker g$ ,  $x \mapsto f(x)$ , whose kernel is  $\ker f$ . The cokernel inequality is the finite-dimensional rank inequality. Terminal-generic dimensions equal the tail dimensions.  $\square$

|                          |                                          |
|--------------------------|------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-015 |
| <b>Role / status</b>     | toy_bridge_theorem / proved              |
| <b>Source bindings</b>   | CM-v19-036                               |
| <b>Comparison policy</b> | not_applicable                           |
| <b>Failure routes</b>    | FR-MISSING, FR-T4, FR-T5                 |

**Remark 8.4** (No factorwise converse). *Cleanliness of  $gf$  does not imply cleanliness of  $f$  or  $g$ . Composition may kill or bypass a defect.*

**Theorem 8.5** (Short-exact-diagram toy bridge). *Consider a commutative diagram in  $\text{Pers}_k^{\text{cons}}$  with exact rows*

$$\begin{array}{ccccccc} 0 & \rightarrow & M' & \rightarrow & M & \rightarrow & M'' \rightarrow 0 \\ & & \downarrow \phi' & & \downarrow \phi & & \downarrow \phi'' \\ 0 & \rightarrow & N' & \rightarrow & N & \rightarrow & N'' \rightarrow 0. \end{array}$$

*Assume all six modules and the three vertical defect profiles are terminal-tame on one common scope. Then*

$$\mu(\phi) \leq \mu(\phi') + \mu(\phi''), \quad \nu(\phi) \leq \nu(\phi') + \nu(\phi'').$$

*Thus cleanliness of the two outer comparisons implies cleanliness of the middle comparison.*

*Proof.* The snake lemma gives an exact sequence from the three kernels and cokernels. Passing to the common stabilized tail and taking dimensions yields the inequalities.  $\square$

|                          |                                              |
|--------------------------|----------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-016     |
| <b>Role / status</b>     | conditional_toy_bridge_theorem / conditional |
| <b>Source bindings</b>   | CM-v19-037                                   |
| <b>Comparison policy</b> | not_applicable                               |
| <b>Failure routes</b>    | FR-MISSING, FR-T4, FR-T5                     |

## 9 Single-Morphism Metric Bridge

**Definition 9.1** (Finite-shift triviality). *For  $\epsilon \geq 0$ , a persistence module  $E$  is  $\epsilon$ -trivial if its translation morphism  $E \rightarrow E(\epsilon)$  is zero.*

**Theorem 9.2** (Single-morphism interleaving bridge). *Let  $M, N$  be pointwise finite-dimensional one-parameter persistence modules and let  $\delta \geq 0$ . Then  $M$  and  $N$  are  $\delta$ -interleaved if and only if there exists a morphism*

$$f : M \longrightarrow N(\delta)$$

*whose kernel and cokernel are both  $2\delta$ -trivial.*

*Proof.* This is the single-morphism characterization consumed by CM-v19-038, with its pointwise finite-dimensional and one-parameter hypotheses retained verbatim in scope.  $\square$

|                          |                                                                 |
|--------------------------|-----------------------------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-017                        |
| <b>Role / status</b>     | toy_bridge_theorem / proved                                     |
| <b>Source bindings</b>   | CM-v19-038; Bauer–Lesnick (2015), Corollary 6.6                 |
| <b>Comparison policy</b> | not_applicable to hard deletion; an interleaving statement only |
| <b>Failure routes</b>    | FR-MISSING, FR-T3, FR-T5                                        |

**Theorem 9.3** (Finite-shift defect implies terminal-generic cleanliness). *Let  $f : M \rightarrow N$  be an actual morphism whose kernel and cokernel are interval-decomposable, terminal-tame, and  $\epsilon$ -trivial for a finite  $\epsilon$ . Then*

$$\mu(f) = \nu(f) = 0.$$

*Consequently, a single-morphism interleaving witness from the preceding theorem is terminal-generic clean when used as the declared Type IV comparison artifact on the same valid scope.*

*Proof.* A terminal-cofinal interval summand has nonzero structure maps across every finite shift sufficiently far out, so it cannot occur in a finite-shift-trivial module. Hence neither defect profile has a terminal-cofinal summand.  $\square$

|                          |                                              |
|--------------------------|----------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-018     |
| <b>Role / status</b>     | conditional_toy_bridge_theorem / conditional |
| <b>Source bindings</b>   | CM-v19-039                                   |
| <b>Comparison policy</b> | not_applicable                               |
| <b>Failure routes</b>    | FR-MISSING, FR-T4, FR-T5                     |

**Theorem 9.4** (Cleanliness does not imply quantitative interleaving). *For every  $\delta > 0$ , terminal-generic cleanliness of an actual comparison morphism does not imply that its kernel and cokernel are  $2\delta$ -trivial and does not imply a  $\delta$ -interleaving.*

*Proof.* Choose  $L > 2\delta$ , let  $M = 0$ ,  $N = k_{[0,L)}$ , and take the zero morphism  $f : M \rightarrow N(\delta)$ . Its kernel is zero and its cokernel has no terminal-cofinal bar, so  $\mu(f) = \nu(f) = 0$ . But the cokernel is not  $2\delta$ -trivial and  $d_I(M, N) = L/2 > \delta$ .  $\square$

|                          |                                          |
|--------------------------|------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-019 |
| <b>Role / status</b>     | toy_separation_theorem / proved          |
| <b>Source bindings</b>   | CM-v19-040                               |
| <b>Comparison policy</b> | not_applicable                           |
| <b>Failure routes</b>    | FR-MISSING, FR-T5                        |

## 10 Controlled Diagnostic Defect and Zero Defect

**Definition 10.1** (Diagnostic dimension budget). *A diagnostic dimension budget is a pair of nonnegative integer bounds*

$$b_{i,W,\tau}^\mu, \quad b_{i,W,\tau}^\nu$$

*satisfying*

$$\mu_{i,W,\tau} \leq b_{i,W,\tau}^\mu, \quad \nu_{i,W,\tau} \leq b_{i,W,\tau}^\nu.$$

*These are bounds on terminal-generic dimensions. They are not bottleneck distances and are not elements of the metric quantale unless an independent typed conversion theorem is supplied.*

**Proposition 10.2** (Bounded diagnostic defect is controlled, not clean). *Finite diagnostic dimension budgets imply controlled terminal-generic defect. They do not imply*

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

*unless every relevant zero conclusion is separately established.*

*Proof.* A positive finite upper bound permits a positive finite diagnostic. Boundedness and vanishing are different predicates.  $\square$

|                          |                                          |
|--------------------------|------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-020 |
| <b>Role / status</b>     | toy_controlled_defect_theorem / proved   |
| <b>Source bindings</b>   | CM-v19-034-037; legacy TB-7 repaired     |
| <b>Comparison policy</b> | not_applicable                           |
| <b>Failure routes</b>    | FR-MISSING, FR-T4, FR-T5                 |

**Proposition 10.3** (Zero diagnostic budget implies cleanliness). *If every invoked valid degreewise diagnostic satisfies*

$$b_{i,W,\tau}^\mu = b_{i,W,\tau}^\nu = 0,$$

*then the status is `certified_zero` and*

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0).$$

*Proof.* The diagnostics are nonnegative integers bounded above by zero. They therefore vanish degreewise and after finite aggregation.  $\square$

|                          |                                              |
|--------------------------|----------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-021     |
| <b>Role / status</b>     | conditional_toy_bridge_theorem / conditional |
| <b>Source bindings</b>   | CM-v19-023                                   |
| <b>Comparison policy</b> | not_applicable                               |
| <b>Failure routes</b>    | FR-MISSING, FR-T4, FR-T5                     |

**Policy 10.4** (Rejection of the legacy mixed ledger). *The legacy expression*

$$\delta^\mu + \delta^\nu + \delta^{\text{real}} + \delta^{\text{apex}} < \text{gap}_\tau$$

is not a well-typed v19 statement when the first two terms are dimension bounds and the latter terms are metric discrepancies. TB-v19 replaces it by two independent records:

1. a diagnostic-dimension record controlling  $\mu$  and  $\nu$ ;
2. a metric ledger  $\mathfrak{d}_{\text{met}} \in V$  with scalarization  $\text{val}_B(\mathfrak{d}_{\text{met}})$  controlling an actual bottleneck discrepancy.

Neither record repairs a missing actual morphism, apex agreement, terminal-tameness certificate, realization, or artifact identity.

|                          |                                                  |
|--------------------------|--------------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-022         |
| <b>Role / status</b>     | metric_repair_policy / operational               |
| <b>Source bindings</b>   | Main typed scalarization; CM-v19-013, CM-v19-028 |
| <b>Comparison policy</b> | runtime_resolved                                 |
| <b>Failure routes</b>    | FR-MISSING, FR-T3, FR-T5                         |

## 11 Metric-Gap-Safe Toy Comparison

**Theorem 11.1** (Sharp two-sided metric toy bridge). *Let  $B, C$  be finite barcodes, possibly with infinite bars, and let  $\tau > 0$ ,  $\varepsilon \geq 0$ . Suppose*

$$d_B(B, C) \leq \varepsilon$$

*and no finite bar of either barcode has length in the closed band*

$$[\tau - \varepsilon, \tau + \varepsilon].$$

*Then*

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon.$$

*Proof.* A bottleneck matching of cost at most  $\varepsilon$  cannot match a retained finite bar on one side to a deleted finite bar on the other under the two-sided closed-band exclusion. Infinite bars match only infinite bars and are retained. Restricting the matching gives the asserted bound.  $\square$

|                          |                                                                |
|--------------------------|----------------------------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-023                       |
| <b>Role / status</b>     | metric_repair_toy_theorem / conditional                        |
| <b>Source bindings</b>   | CM-v19-012                                                     |
| <b>Comparison policy</b> | fixed_metric_gap_certified; runtime mode: metric_gap_certified |
| <b>Failure routes</b>    | FR-MISSING, FR-T3, FR-T5                                       |

**Theorem 11.2** (Scalarized metric-ledger toy audit). *Let  $(V, \oplus, 0_V, \preceq)$  be the declared metric quantale and*

$$\text{val}_B : V \rightarrow [0, +\infty]$$

*a monotone subadditive scalarization. Let  $\mathfrak{d}_{\text{met}}$  be the aggregate metric ledger element for a comparison of the windowed pre-threshold profiles  $B$  and  $C$ . Suppose the evidence bundle proves*

$$d_B(B, C) \leq \varepsilon_B := \text{val}_B(\mathfrak{d}_{\text{met}})$$

*and*

$$\varepsilon_B < \text{gap}_\tau(B, C) := \min\{\text{clear}_\tau(B), \text{clear}_\tau(C)\}.$$

*Then the post-deletion comparison is metric-gap-certified and*

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon_B.$$

*This certifies the stated metric transport only. It does not establish Type IV cleanliness, theorem truth, or any missing non-scalar obligation.*

*Proof.* The strict inequality gives the required two-sided avoidance of the closed threshold band. Apply the sharp metric toy bridge. The final sentence follows from the v19 separation of metric and non-scalar obligations.  $\square$

|                          |                                                                |
|--------------------------|----------------------------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-024                       |
| <b>Role / status</b>     | conditional_metric_toy_bridge_theorem / conditional            |
| <b>Source bindings</b>   | CM-v19-013, CM-v19-028                                         |
| <b>Comparison policy</b> | fixed_metric_gap_certified; runtime mode: metric_gap_certified |
| <b>Failure routes</b>    | FR-MISSING, FR-T3, FR-T5                                       |

**Proposition 11.3** (Exact comparison and metric robustness are distinct). *An exact equality or profile isomorphism may certify a specific exact comparison without any positive metric reserve. It does not by itself certify robustness under positive perturbation. Conversely, a metric-gap-certified comparison does not manufacture an exact isomorphism.*

*Proof.* The two runtime modes have different evidence and conclusions. A profile can be exactly compared with itself even when a bar lies exactly at threshold, where positive gap is zero. A finite bottleneck bound does not imply isomorphism.  $\square$

|                          |                                          |
|--------------------------|------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-025 |
| <b>Role / status</b>     | toy_separation_theorem / proved          |
| <b>Source bindings</b>   | CM-v19-014, CM-v19-015                   |
| <b>Comparison policy</b> | runtime_resolved                         |
| <b>Failure routes</b>    | FR-MISSING, FR-T3, FR-T5                 |

## 12 Combined Audit Readiness Without Compensation

**Definition 12.1** (Toy bridge obligation bundle). *A toy bridge obligation bundle contains independent atoms for:*

1. *source-category and source-theorem validity;*
2. *Core readability and constructibility;*
3. *comparison mode and comparison evidence;*
4. *actual tower or representative artifacts when invoked;*
5. *terminal tameness and monitored-scope validity;*
6. *metric ledger and threshold-gap records when metric mode is invoked;*
7. *diagnostic records when Type IV is invoked;*
8. *claim status, artifact identity, repair ancestry, and manifest consistency.*

**Theorem 12.2** (Non-compensating toy readiness). *A toy bridge record is audit-ready only when every required obligation atom passes and every excluded atom has justified `not_invoked` status. A passing metric atom cannot compensate for an undefined actual-morphism atom; a clean Type IV atom cannot compensate for a failed threshold-gap atom; and a complete manifest cannot compensate for a false theorem hypothesis.*

*Proof.* The atomic status calculus aggregates required obligations by dependency closure and non-compensation. Undefined, rejected, and not-invoked states retain their own semantics and are not numerically canceled by passing siblings.  $\square$

|                          |                                                     |
|--------------------------|-----------------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-026            |
| <b>Role / status</b>     | operational_theorem / proved within audit semantics |
| <b>Source bindings</b>   | CM-v19-028, CM-v19-029; Foundation Appendix F       |
| <b>Comparison policy</b> | runtime_resolved                                    |
| <b>Failure routes</b>    | FR-MISSING, FR-T1, FR-T2, FR-T3, FR-T4, FR-T5       |

## 13 Toy Bridge Manifest and Local Claim Register

**Definition 13.1** (Canonical toy bridge manifest). *A TB-v19 manifest records at minimum:*

1. *immutable RunID, ManifestID, SchemaID, theory version, CR version, TB revision, and artifact root;*
2. *toy source category, source theorem, tower levels, transitions, and apex;*
3. *realization route, constructibility evidence, monitored degrees, windows, and thresholds;*
4. *window-first repaired evaluations and any representative or eligibility records;*
5. *claim-level comparison policy and resolved runtime mode;*
6. *actual comparison artifacts and terminal-tameness evidence when Type IV is invoked;*
7. *degreewise and total  $\mu/\nu$  diagnostics only in valid invoked cases;*
8. *metric ledger, scalarization, actual discrepancy evidence, and two-sided threshold-gap record when metric mode is invoked;*
9. *every obligation atom, dependency, blocker, status, and failure route;*



10. local TB UID, source CM/CR UIDs, permitted use, forbidden reading, and repair ancestry;
11. immutable artifact identities, hashes, replay information, and final manifest/readiness status;
12. explicit external-domain non-claim boundary.

**Theorem 13.2** (Manifest necessity for reusable toy bridges). *A TB theorem may be consumed by a later Bridge Program only if its manifest is complete, consistent, replayable at the declared scope, and source-bound to the exact theorem revision used. Manifest completeness is necessary for auditable reuse but is not sufficient for mathematical truth.*

*Proof.* Without the listed fields, the later consumer cannot reconstruct source semantics, hypotheses, comparison mode, artifacts, diagnostics, or status. Conversely, serialization of false or unsupported data does not make the theorem true.  $\square$

|                          |                                                       |
|--------------------------|-------------------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-027              |
| <b>Role / status</b>     | operational_theorem / operational                     |
| <b>Source bindings</b>   | Foundation Appendix G; CR-G-002, CR-G-010; CM-v19-041 |
| <b>Comparison policy</b> | not_applicable                                        |
| <b>Failure routes</b>    | FR-MISSING, FR-T5                                     |

**Definition 13.3** (Local TB UID rule). *Every theorem, policy, separation result, and non-claim in this revision has a globally namespaced UID of the form*

*ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-NNN.*

*The short identifier is display-only. A later CR revision imports these UIDs without mutating their source text or verdict.*

**Remark 13.4** (Complete local register display). *The following table is the canonical human-readable display of all 31 local records in this revision. It enumerates TB-v19-001 through TB-v19-031 without gaps, including definitions, corollaries, conditional theorems, operational rules, non-claim records, and the package-closure theorem. The machine-readable JSON and CSV registers contain the same 31 identifiers.*

| Short ID   | Registered content                                    | Evidence status |
|------------|-------------------------------------------------------|-----------------|
| TB-v19-001 | Toy bridge sovereignty and external-domain boundary   | operational     |
| TB-v19-002 | Canonical comparison-mode trichotomy for toy bridges  | operational     |
| TB-v19-003 | Finite barcode readability with actual-morphism guard | proved          |
| TB-v19-004 | Eventual-constancy toy bridge                         | conditional     |
| TB-v19-005 | Finite barcode-profile stabilization                  | conditional     |
| TB-v19-006 | Exact-conjugacy invariance of toy diagnostics         | proved          |
| TB-v19-007 | Finite-dimensional monotone stabilization             | proved          |
| TB-v19-008 | Vector-space stabilization bridge                     | conditional     |
| TB-v19-009 | Finite-length monotone stabilization                  | proved          |
| TB-v19-010 | Finite-length module stabilization bridge             | conditional     |

| Short ID   | Registered content                                                       | Evidence status |
|------------|--------------------------------------------------------------------------|-----------------|
| TB-v19-011 | Eligible one-way PH1-to-Ext1 toy edge                                    | conditional     |
| TB-v19-012 | Representative invariance of eligible toy Ext discharge                  | conditional     |
| TB-v19-013 | Terminal-generic index identity                                          | proved          |
| TB-v19-014 | Finite direct-sum additivity                                             | proved          |
| TB-v19-015 | Composition-controlled toy bridge                                        | proved          |
| TB-v19-016 | Short-exact-diagram toy bridge                                           | conditional     |
| TB-v19-017 | Single-morphism interleaving bridge                                      | proved          |
| TB-v19-018 | Finite-shift defect implies terminal-generic cleanliness                 | conditional     |
| TB-v19-019 | Terminal-generic cleanliness does not imply<br>quantitative interleaving | proved          |
| TB-v19-020 | Bounded diagnostic defect is controlled, not clean                       | proved          |
| TB-v19-021 | Zero diagnostic budget implies cleanliness                               | conditional     |
| TB-v19-022 | Rejection of legacy mixed diagnostic/metric ledger                       | operational     |
| TB-v19-023 | Sharp two-sided metric toy bridge                                        | conditional     |
| TB-v19-024 | Scalarized metric-ledger toy audit                                       | conditional     |
| TB-v19-025 | Exact comparison and metric robustness are distinct                      | proved          |
| TB-v19-026 | Non-compensating toy readiness                                           | proved          |
| TB-v19-027 | Manifest necessity for reusable toy bridges                              | operational     |
| TB-v19-028 | Iwasawa-style pattern only                                               | spec            |
| TB-v19-029 | Bounded-control caution only                                             | proved          |
| TB-v19-030 | Toy dissipative instance                                                 | conditional     |
| TB-v19-031 | TB-v19 package closure                                                   | proved          |

## 14 Legacy TB Migration

**Policy 14.1** (Immutable migration). *The v18.3 Candidate TB artifact is not overwritten. Its statements are mapped to this revision as preserved, strengthened, split, deprecated, or rejected. Any substantive repair creates a new UID.*

| Legacy ID | Legacy content                  | v19 disposition    | Canonical destination                                     |
|-----------|---------------------------------|--------------------|-----------------------------------------------------------|
| TB-1      | finite barcode<br>readability   | strengthened       | TB-v19-003;<br>actual-morphism guard<br>added             |
| TB-2      | eventual constant tower         | strengthened       | TB-v19-004; colimit, apex,<br>artifact, tameness explicit |
| TB-3      | finite barcode<br>stabilization | strengthened       | TB-v19-005; realized<br>persistence tower required        |
| TB-4      | vector-space bridge             | split/strengthened | TB-v19-007–008                                            |
| TB-5      | finite-length module<br>bridge  | split/strengthened | TB-v19-009–010                                            |
| TB-6      | zero defect budget              | strengthened       | TB-v19-021                                                |
| TB-7      | bounded defect not<br>clean     | preserved/typed    | TB-v19-020                                                |

| Legacy ID | Legacy content                 | v19 disposition                            | Canonical destination                               |
|-----------|--------------------------------|--------------------------------------------|-----------------------------------------------------|
| TB-8      | mixed ledger controlled bridge | rejected as ill-typed                      | TB-v19-022 plus TB-v19-024/026                      |
| TB-9      | ledger-safe zero defect        | split                                      | diagnostic zero TB-v19-021; metric audit TB-v19-024 |
| TB-10     | manifest completeness          | strengthened                               | TB-v19-027                                          |
| TB-11–12  | arithmetic readiness           | demoted to non-claim pattern               | Section 15                                          |
| TB-13     | dissipative toy stabilization  | retained only as an instance of TB-v19-004 | Section 15                                          |

**Remark 14.2** (Principal semantic repair). *The most important migration is TB-8/TB-9. Diagnostic dimensions and bottleneck discrepancies no longer share an untyped scalar sum. This removes a category error and prevents a positive threshold reserve from being read as a proof of zero Type IV defect.*

## 15 Arithmetic and PDE Readiness as Explicit Non-Claims

**Proposition 15.1** (Iwasawa-style pattern only). *The implication*

$$\begin{array}{c} \text{eventual repaired-profile} \\ \text{stabilization} \end{array} \implies \text{exact tower artifact} \implies (\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

*is a valid toy pattern. It is not an Iwasawa theorem and does not establish information preservation along an Iwasawa tower.*

*Proof.* The toy implication is TB-v19-004. An Iwasawa application requires an arithmetic source category, purpose-faithful realization, finite-stage-to-limit transport, actual comparison artifact, and a proof that the realization preserves the arithmetic information of interest.  $\square$

|                          |                                          |
|--------------------------|------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-028 |
| <b>Role / status</b>     | non_claim_pattern / spec                 |
| <b>Source bindings</b>   | TB-v19-004; CR-G-001, CR-G-004, CR-G-005 |
| <b>Comparison policy</b> | not_applicable                           |
| <b>Failure routes</b>    | FR-NONVERDICT, FR-T5                     |

**Proposition 15.2** (Bounded-control caution only). *A uniformly bounded family of toy terminal-generic defects is controlled but not necessarily clean. Therefore a future arithmetic control theorem with bounded finite-layer error cannot be interpreted as information-loss-free without an additional vanishing or finite-shift-triviality theorem.*

*Proof.* Apply TB-v19-020 and compare with TB-v19-018. Bounded dimension and finite-shift triviality are different hypotheses.  $\square$

|                          |                                                  |
|--------------------------|--------------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-029         |
| <b>Role / status</b>     | non_claim_caution / proved as logical separation |
| <b>Source bindings</b>   | TB-v19-018, TB-v19-020                           |
| <b>Comparison policy</b> | not_applicable                                   |
| <b>Failure routes</b>    | FR-NONVERDICT, FR-T5                             |

**Proposition 15.3** (Toy dissipative instance). *A finite-dimensional toy dissipative tower whose repaired profiles are eventually constant, with declared apex agreement and exact tower artifact, has clean terminal-generic diagnostics.*

*Proof.* The dissipative label contributes no proof step. The conclusion is exactly TB-v19-004 applied to the repaired profile tower.  $\square$

|                          |                                              |
|--------------------------|----------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-030     |
| <b>Role / status</b>     | conditional_toy_bridge_theorem / conditional |
| <b>Source bindings</b>   | TB-v19-004                                   |
| <b>Comparison policy</b> | fixed_exact; runtime mode: exact             |
| <b>Failure routes</b>    | FR-MISSING, FR-T4, FR-T5                     |

**Remark 15.4** (No PDE theorem). *The preceding proposition is not an energy inequality, compactness theorem, weak-solution theorem, uniqueness theorem, regularity theorem, blow-up exclusion, or Navier–Stokes result.*

## 16 Explicit Exclusions

No statement in this appendix asserts any of the following:

- a global exact or functorial hard-threshold deletion on all persistence morphisms;
- global non-expansiveness of hard positive-threshold deletion;
- unconditional crop–collapse commutation;
- an automatic tower comparison artifact;
- a numerical diagnostic when Type IV is `undefined` or `not_invoked`;
- equivalence between bounded diagnostic dimension and zero diagnostic dimension;
- equivalence between terminal-generic cleanliness and quantitative interleaving;
- conversion of diagnostic dimension into bottleneck distance without a typed theorem;
- conversion of metric slack into an actual morphism, eligibility record, terminal-tameness certificate, or proof;
- reverse  $\text{Ext}^1$ -to- $\text{PH}_1$  transport or global  $\text{PH}_1 = \text{Ext}^1$ ;
- identification of  $\mu_{\text{Tw}}$  with  $\mu_{\text{Collapse}}$ ;
- information-loss-free projection along an arithmetic infinite tower;
- any external-domain conclusion from a Core pass;
- any theorem status from Search, AI, Companion, Spec, or manifest completeness alone.

## 17 Closure, Provenance, and Status

**Theorem 17.1** (TB-v19 package closure). *Within the declared toy source regimes, Appendix TB-v19.0.0-r1 provides:*

1. *exact eventual-constancy bridge theorems with actual colimit-to-apex artifacts;*
2. *source-side stabilization theorems for finite-dimensional vector spaces and finite-length modules;*
3. *a conditional one-way eligible degree-one toy edge;*
4. *exact-conjugacy, index, finite-sum, composition, and short-exact-diagram defect calculus;*
5. *a single-morphism interleaving bridge, finite-shift-to-cleanliness implication, and strict converse failure;*
6. *a typed separation between diagnostic dimension budgets and metric ledgers;*
7. *sharp metric-gap transport under hard deletion;*
8. *non-compensating obligation and manifest rules;*
9. *immutable legacy migration and explicit external-domain non-claims.*

*No stronger conclusion follows without an additional theorem whose hypotheses and source semantics are separately registered.*

*Proof.* Each item is established in the cited section from the displayed hypotheses and immutable source bindings. The last sentence follows from the claim taxonomy, weaker-safe interpretation, and the explicit non-claim boundary.  $\square$

|                          |                                                              |
|--------------------------|--------------------------------------------------------------|
| <b>TB UID</b>            | ak-hpdst:19.0.0:TB-v19.0.0-r1:TB-v19-031                     |
| <b>Role / status</b>     | toy_bridge_package_theorem / proved at registered strengths  |
| <b>Source bindings</b>   | TB-v19-001-030; CM-v19-031-041; Foundation F-G               |
| <b>Comparison policy</b> | mixed; resolved per consumed claim                           |
| <b>Failure routes</b>    | FR-MISSING, FR-T1, FR-T2, FR-T3, FR-T4, FR-T5, FR-NONVERDICT |

| Component                                            | Status in TB-v19.0.0-r1                      |
|------------------------------------------------------|----------------------------------------------|
| Toy source definitions                               | interface definitions, non-existence-neutral |
| Eventual-constancy bridge                            | conditional toy bridge theorem               |
| Vector-space stabilization                           | proved toy source theorem                    |
| Finite-length stabilization                          | proved toy source theorem                    |
| Eligible $\text{PH}_1 \rightarrow \text{Ext}^1$ edge | conditional one-way toy bridge               |
| Exact defect calculus                                | proved / conditional at CM-r2 strength       |
| Single-morphism bridge                               | proved in pfd one-parameter scope            |
| Bounded diagnostic control                           | proved separation theorem                    |
| Metric-gap transport                                 | conditional metric-repair theorem            |
| Manifest and status algebra                          | operational governance                       |
| Arithmetic/PDE readiness                             | pattern or caution; external non-claim       |

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| Component       | Status in TB-v19.0.0-r1               |
|-----------------|---------------------------------------|
| CR-r3 promotion | pending immutable governance revision |

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## References

- [1] U. Bauer and M. Lesnick, *Induced matchings and the algebraic stability of persistence barcodes*, Journal of Computational Geometry 6(2) (2015), 162–191.